

# OsteoTrack: Early Detection of Fracture Non-Union Through Non-Invasive Bone Healing Monitoring

## Group D: MPHY0033 Final Report

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Revision 0.0

### Notes for Ziyan

- Formatting and verb tense will be synchronised for the final submission
- References will be properly integrated for the final submission.
- Word count (excluding references and appendices) will be reduced to 8000 words.
- Redundancies between sections will be eliminated where possible

### Questions for Ziyan

- Is the discussion of the AI integration sufficient? Is more required on the clinical background, existing methods, and justification of the use of an AI model?
- Are we discussing this device as if we were proposing it, or that it's something we've made and this is our plan for it?
- Our business plan (appendix 11.2) focuses on long term growth, estimating up to 4000 devices in use. Should we taper our plan back down to the manufacturing and deployment of 100 devices?

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# 1 Executive Summary

Fracture non-union presents approximately 330,000 times per annum in the United Kingdom, costing the National Health Service (NHS) up to £79,000 per patient. Early diagnosis and intervention of these cases can reduce the burden patients place on NHS resources, the workload of clinical specialists, and long-term negative health outcomes of patients. Existing non-union early diagnosis requires up to 5 in-person follow-up appointments with orthopaedic surgeons and x-ray imaging, exposing patients to ionising radiation and costing the NHS £56 per scan. Limited resources, patient risk, and inconvenient scheduling limit the number patients referred to these follow-up appointments, however many patients recommended self-management eventually require an in-person appointment and x-rays. OsteoTrack provides non-ionising and non-invasive bone healing monitoring from community points of care, allowing more high-risk non-union cases to be identified with less resource use.

The device, resembling a blood pressure cuff, contains a motor that vibrates a patient's limb at a range of frequencies. Vibrational activity within the limb is captured by accelerometers and converted to a cross spectrum capturing the resonance frequency of bone, a clinically validated indicator of bone healing progression. The spectrum is provided to a time aware long short-term memory artificial intelligence model, which observes trends in vibrational activity between scans to determine if bone healing is following an expected trajectory. Patients deviating from such trajectories are predicted to be high-risk for non-union, allowing clinicians to make an early diagnosis and ensure timely intervention.

The device and associated AI algorithm itself do not make a clinical diagnosis; this remains the responsibility of clinicians. The device is designed for Class IIa compliance under MHRA guidelines, with clinical trials planned to validate performance and support regulatory approval.

The device is marketed to the NHS directly due to the stability of its procurement model and its shift towards digital and preventative health. Additionally, a 24.4% compound annual growth rate for AI in healthcare demonstrates favourable conditions for the market, and limited competition offers larger capturable market share. To reduce barriers to adoption and increase early-stage company income, a per-patient pricing model is used. Each patient monitored is priced at £35, a substantial saving from the cost of multiple x-ray scans during current follow-up methods. Marketing strategies are prepared to target key NHS decision-makers to build trust and expedite procurement contracts.

The first two years of development focus on research and development, clinical trials, and regulatory approval, and are expected to cost the company £315,000. Years 3-7 involve the production and clinical integration of the device, with 100 units produced at a cost of £40,117. Long term growth with additional devices projects over £1 million in net income per year, accounting for materials and labour, regulatory commitments, and operational overhead. A range of applicable grants offer additional funding opportunities, reducing dependence on debt and equity dilution and allowing for more sustainable growth of OsteoTrack.



## 2 Clinical Background

Fractures are a common reason for presenting to hospital around the world. Hospital admission to the National Health Service (NHS) for fracture occurs annually at a rate of 47.84 per 10,000 head of population<sup>1</sup>, meaning that approximately 330,000 presentations occur annually across the UK. Fracture non-union is a complication in bone healing that can stem from range of issues, including inadequate immobilisation, poor blood flow, and comorbidities. While difficult to prevent, timely diagnosis can ensure rapid intervention and prevent additional bone damage or muscle atrophy. Early treatment of non-union also mitigates chronic conditions such as ongoing pain, limited range of motion, and poor mental health that impact patient productivity and drain NHS resources [1].

As of 2017, the risk of non-union per fracture is 1.9%, but can be as high as 9% depending on the impacted bone and patient demographic [2]. Given these statistics, rigorous patient follow-up is required to ensure patients with improper bone healing can be identified and potential non-unions addressed. The current patient flow upon presenting with a possible fracture is as follows [3]:

1. Patient presents at an urgent care facility. Fracture is diagnosed by imaging (typically with X-ray, though CT may be required)
2. Patient is referred to the Orthopaedics team to determine management (i.e. type of immobilisation, determination on whether surgery is required etc.)
3. Admission for surgery, if necessary
4. Patient discharged with referral to a virtual fracture clinic (VFC) which decides between additional face-to-face follow-ups and self-management
5. If additional follow-up required, attend appointment with an orthopaedic surgeon typically 1-2 weeks later. This may be virtual for simple, stable fractures. If in person, this includes X-ray imaging
6. Ongoing monitoring with repeated fracture clinic appointments (either virtual or in person), with X-ray imaging typically undertaken at each in-person appointment.
7. Final discharge

Cases which require in-person follow-up to monitor healing progression typically involve up to 5 additional follow-up appointments, each requiring consultation with an orthopaedic surgeon and X-ray imaging, costing the NHS approximately £56 per X-ray<sup>2</sup> and exposing the patients to the cumulated risks of repeated ionising radiation exposure. In addition, 48% of patients whose fractures were initially deemed treatable entirely through a VFC process of follow-up have been shown to ultimately require an in-person appointment<sup>3</sup>. Delayed treatment of non-union not only impacts

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<sup>1</sup> Jennison T, Brinsden M. Fracture admission trends in England over a ten-year period. *Ann R Coll Surg Engl*. 2019 Mar;101(3):208-214.

<sup>2</sup> Evidence review for the clinical and cost effectiveness of imaging during the management of osteoarthritis: Osteoarthritis in over 16s: diagnosis and management: Evidence review M. London: National Institute for Health and Care Excellence (NICE); 2022 Oct. (NICE Guideline, No. 226.)

<sup>3</sup> Jain T, Rahij H, Hilal Z, Anwar R. Orthopaedic Telemedicine Clinics: A Patient Satisfaction Survey of 522 Patients. *Cureus*. 2024 Nov 10;16(11):e73392.

patient outcomes, but also incurs significant healthcare expenditure for the NHS in the range of £7,000 to £79,000<sup>4</sup>. A broadly deployable screening process that can indicate abnormal healing progression early and does not require the use of ionising radiation would be both safer and more convenient for the patient, whilst also relieving the pressure on the NHS and providing significant cost savings; it is exactly this need that OsteoTrack device has been designed to meet.

OsteoTrack device provides a low cost, non-ionising method for monitoring fracture healing progression in primary care settings or hospitals, allowing non-union cases to be recognised early. Designed to be operated by GPs or clinical specialists, it reduces demand on orthopaedic surgeons and radiologists while retaining appropriate clinical supervision. Patients can easily obtain local follow-up care at a GP surgery or clinical centre, as an X-ray facility is not necessary; the increased location options will improve appointment availability, relieving the pressure on fracture clinic staff and freeing up equipment at central fracture clinics. Together, these advantages offer an alternative low-risk follow-up pathway with reduced cost to the NHS, allowing for increased patient referral and reducing the risk of non-union being missed due to premature discharge.

1. Wildemann, B., Ignatius, A., Leung, F. et al. Non-union bone fractures. *Nat Rev Dis Primers* 7, 57 (2021). <https://doi.org/10.1038/s41572-021-00289-8>
2. Mills, L.A. and Simpson, A.H.R.W. (2013) “The relative incidence of fracture non-union in the Scottish population (5.17 million): a 5-year epidemiological study,” *BMJ Open*, 3(2), p. e002276. Available at: <https://doi.org/10.1136/bmjopen-2012-002276>.
3. <https://www.kch.nhs.uk/services/services-a-to-z/fracture-care-team/>

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<sup>4</sup> Mills, L.A. and Simpson, A.H.R.W. (2013) “The relative incidence of fracture non-union in the Scottish population (5.17 million): a 5-year epidemiological study,” *BMJ Open*, 3(2), p. e002276. Available at: <https://doi.org/10.1136/bmjopen-2012-002276>; Sibanda, V. et al. (2022) “Outcomes of the Treatment of Fracture Non-union Using Combined Magnetic Field Bone Growth Stimulation: Experiences From a UK Trauma Unit,” *Cureus*, 14(5), p. e25100. Available at: <https://doi.org/10.7759/cureus.25100>.

## 3 Design

### 3.1 Design Overview

The device we proposed is a portable, non-invasive tool for monitoring the healing of non-union fractures in long bones, including the femur, tibia, fibula, humerus, ulna, and radius, by measuring how the bone vibrates non-invasively, with no radiation or surgery involved. It is purely diagnostic: it provides a qualitative measure of how well a fracture is healing over time and does not deliver any therapeutic effect. It has three main parts, all built into a modified blood pressure cuff:

- Excitation system: a motorised assembly that delivers a controlled frequency sweep vibration to the bone through the skin surface.
- Sensing system: two accelerometers that record the vibrational response.
- Interface system: an embedded circuit board that digitises the raw signals and transmits the full frequency spectrum wirelessly to a clinical application, whereupon an Artificial Intelligence (AI) model interprets the data.

The device does not produce a healing result on its own. Each measurement produces a graph of vibration amplitude against frequency, sent wirelessly to a clinical app where an AI model tracks changes week by week. A blood pressure cuff was chosen as the housing because inflating it simultaneously holds the contact disc and accelerometers firmly against the limb, compresses the soft tissue to reduce signal damping, and accommodates different limb sizes, all in a single automated step.

### 3.2 Basis for Design

The design is based on a 2022 clinical study by Vien et al., carried out in Melbourne [1]. The physical principle is that a fracture introduces a discontinuity in the bone that increases its local flexibility and reduces its bending stiffness. This reduction in stiffness lowers the bone's natural frequency, at which it vibrates most strongly under excitation. Nikiforidis et al. [2] demonstrated analytically and experimentally that this natural frequency drops after fracture and recovers progressively as callus forms and stiffness is restored. By tracking this frequency shift non-invasively from week to week, it is therefore possible to monitor healing without X-rays. In principle, however, it is difficult to isolate the resonance frequency of bone from surrounding tissue. Vien et al. [1] tested this on 20 healthy participants using a blood pressure cuff and an impact hammer, and their findings directly informed the design decisions in this device, as summarised in **Error! Reference source not found..**

Table 1. Key findings from Vien et al. [1] and design decisions

| Finding                                                                                                                                                      | Design Decision                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| As bone heals and stiffens, its resonance frequencies shift measurably. The most diagnostically useful frequency change for bone occurs between 60 and 110Hz | Motor sweeps 60-110Hz; AI tracks how the frequency spectrum changes week by week to assess healing progress |
| Even with soft tissue present, the bone-related signal is strong enough to dominate in this frequency range. Bone length alone explains ~43% of the variance | A transcutaneous measurement carries enough bone information for the AI to track healing                    |
| Placing sensors at the femoral epicondyles minimises the soft tissue layer between the sensor and the bone                                                   | Accelerometers positioned at the epicondyles to maximise signal quality                                     |
| Cuff pressure above 120mmHg produces stable measurements                                                                                                     | Automated pump maintains 120-180 mmHg throughout each session                                               |

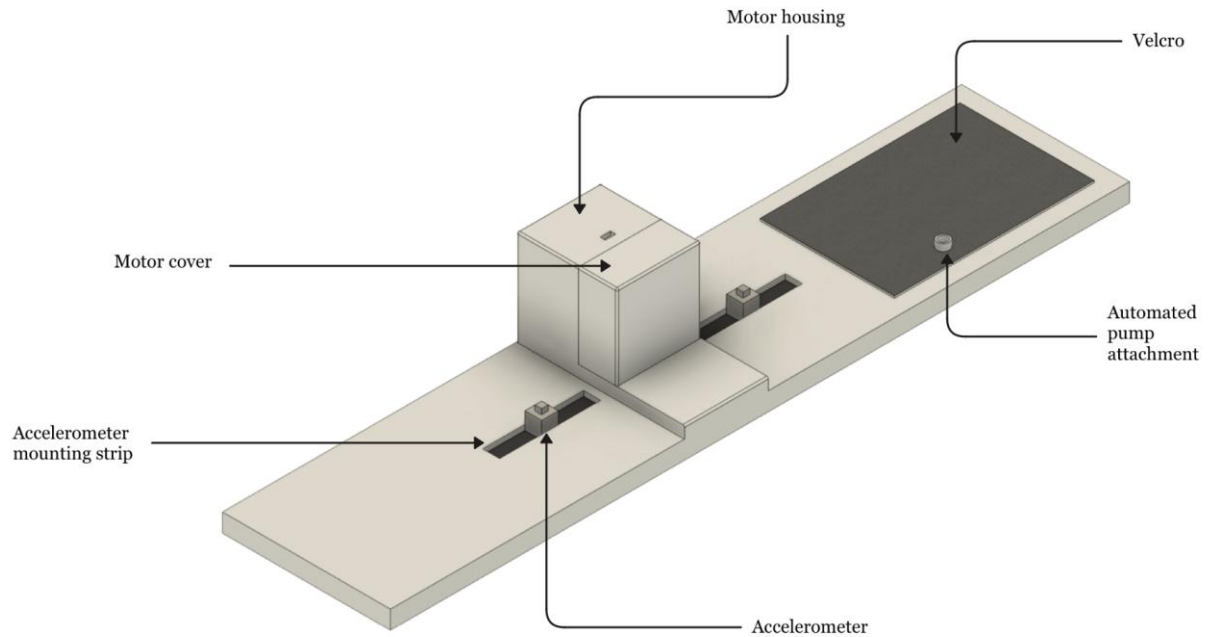
While the Vien et al. [1] study validated the measurement principle, its prototype was not suitable for clinical use, since it required a manually operated impact hammer, a hand-pumped cuff, and a large non-portable analyser. This device replaces each of those with something more practical, as shown in **Error! Reference source not found..**

Table 2. Design changes from Vien et al. [1]

| Feature     | Vien et al. [1]           | This device                                             | Rationale                                                         |
|-------------|---------------------------|---------------------------------------------------------|-------------------------------------------------------------------|
| Excitation  | Manual impact hammer      | Motorised frequency sweep                               | Eliminates operator variability; controlled and repeatable input. |
| Inflation   | Manual bulb               | Automated pump + display                                | Consistent pressure regardless of operator                        |
| Acquisition | Non-portable lab analyser | Wireless Bluetooth modules transmitting to clinical app | Point-of-care use; no external equipment                          |

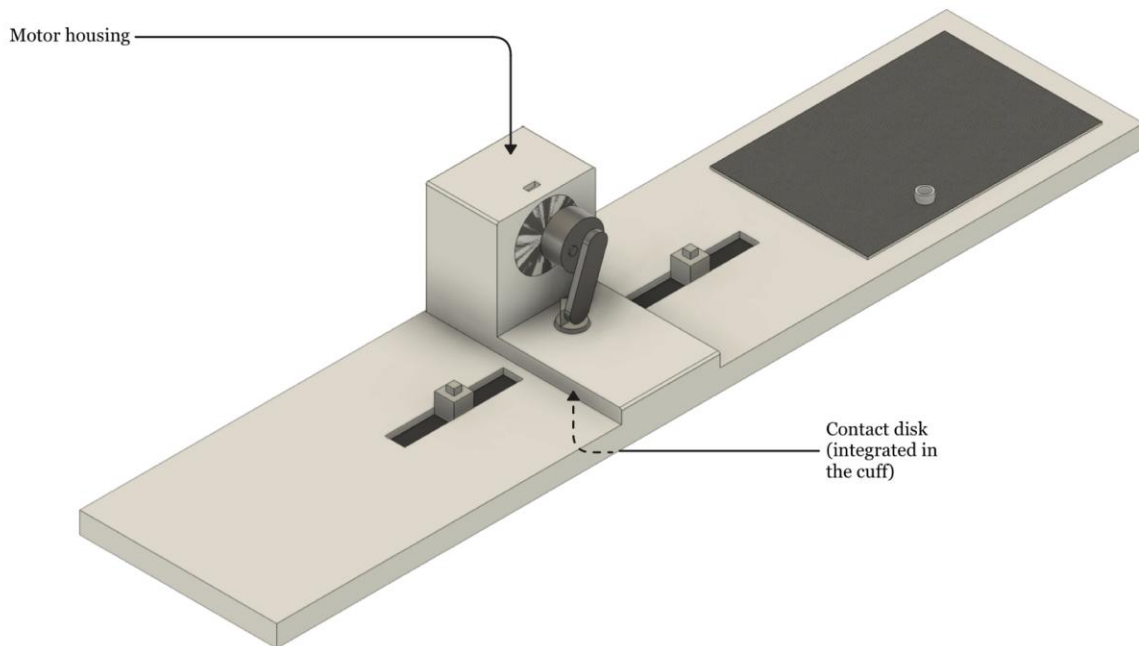
### 3.3 Mechanical Design

Figure 1 shows the complete device as it would appear in clinical use, with the motor housing closed and all components in place.



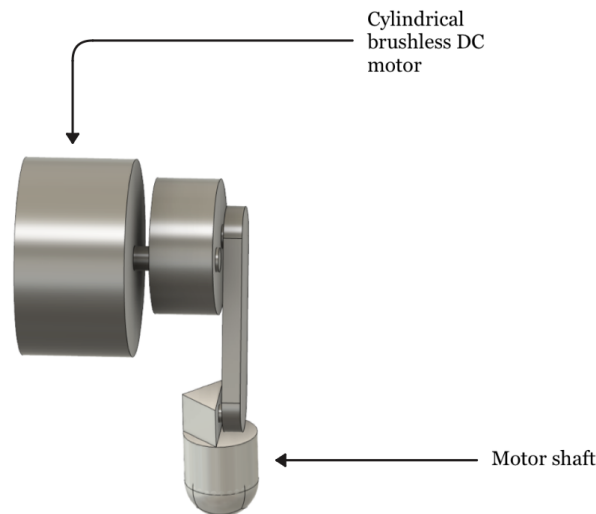
*Figure 1. Complete device assembly with the motor covered*

Figure 2 shows the same assembly with the motor cover removed, revealing the internal layout. The motor sits centrally with the shaft and contact disc pointing downward toward the skin. The accelerometers sit in adjustable slots on either side and can be slid inward or outward to align with the epicondyles of different patients.



*Figure 2. Cuff assembly without the motor cover*

Figure 3 shows the motor and shaft assembly in isolation. As the motor spins, the offset shaft arm converts that rotation into an up-and-down oscillation at the contact disc, which presses against the skin and drives vibration into the bone beneath.



*Figure 3. Motor body with shaft*

## 4 A comprehensive list of components, including their costs, sources, materials, specifications, and design justifications, is provided in Appendixes

Appendix 1: Components.

### 4.1 Electrical and Software implementation

TBD: Software workflow (maybe combined to the other workflow below?)

TBD: Electrical component diagram

### 4.2 Artificial Intelligence Integration

TBD: Figure of frequency plot

TBD: Figure of AI model architecture

Each structure in the limb has a natural frequency, where it will resonate and produce a large signal for the accelerometer. During the frequency sweep, the device applies different frequencies to the limb to capture where structures resonate and where signal is attenuated. The accelerometer signals are then converted to frequency spectrums using a Fourier transform and are combined to form a cross spectrum [3]. The cross spectrum visualises activity seen by both sensors during the sweep and suppresses uncorrelated noise, characterising how the limb behaves to vibrational inputs. The cross spectrum has a range from 60-110 Hz and is discretised in intervals of 1 Hz.

TBD: Paragraph explaining why AI is useful for this application compared to the existing methods

Cross spectrum data can be collected at each follow up appointment and used to predict if the patient is following a healthy bone healing trajectory or is at risk of non-union. A deep learning AI model is used to process large amounts of input data and extract relevant patterns and trends that might be difficult to identify with standard statistical methods or visual inspection.

We propose a time-aware Long Short Term Memory (T-LSTM) model for processing the 50 input values from the cross spectrum. Proposed by Baytas et al. [4], the model is built to accept successive inputs of ordered data that are sampled irregularly. A cross spectrum is recorded at each patient follow-up and provided to the model, which observes trends between the current and past inputs. The model handles the irregular spacing of data recordings by decaying the memory saved within the model; the longer apart two appointments are, the less previous appointments are considered in the model's prediction. There is also no limit on the number of inputs, so both high-risk patients with frequent follow ups as well as lower risk patients can be monitored. The model aims to categorise fracture patients as low risk or high risk for non-union through monitoring changes in the frequency spectrum over each follow up, observing a patient's healing trajectory and comparing it to known healthy trajectories. Patient cross spectra flagged as at high risk of non-union can alert GPs/clinical specialists to prompt consultation with in-person fracture clinics and imaging with X-rays.

Training data will be obtained from clinical trials (discussed in Clinical trials) and integration into fracture clinics, providing frequency response curves from patients experiencing the current standard of care that can be labelled according to whether non-union subsequently developed.

### 4.3 Measurement Procedure

The measurement process is designed to be carried out by any clinical staff member in a few minutes, without specialist training. The steps are as follows:

1. Position the patient with the limb relaxed and fully extended. Place the cuff around the limb at the site of the fracture.
2. Slide the two accelerometer blocks along the cuff strips until they sit directly over the bony landmarks on either side of the joint. The accelerometers should face each other across the limb.
3. Activate the automated pump. The cuff inflates to the target pressure of 120-180 mmHg, pressing the contact disc and accelerometers firmly against the limb and compressing the soft tissue to improve signal quality. The display confirms when the correct pressure is reached.
4. Start the frequency sweep via the companion app. The motor steps through frequencies from 60 Hz to 110 Hz in 1 Hz increments, vibrating the bone at each frequency for approximately 10 seconds before moving to the next step. At each step, the accelerometers record the bone's vibrational response, and the processor averages the signal across multiple cycles to reduce noise.

5. Once the sweep is complete, the device transmits the full amplitude-versus-frequency spectrum wirelessly to the clinical app. The cuff is then deflated and removed. The entire procedure takes approximately ten minutes.
6. The AI model in the app compares this week's spectrum against the patient's previous measurements and the expected healing trajectory from clinical trial data, outputting a healing status for the clinician to review.

#### REFERENCES USED IN THIS SECTION:

[1] Vien, B.S., Chiu, W.K., Russ, M. and Fitzgerald, M. (2022) 'Modal frequencies associations with musculoskeletal components of human legs for extracorporeal bone healing assessment based on a vibration analysis approach', *Sensors*, 22(2), p. 670. <https://doi.org/10.3390/s22020670>

[2] Nikiforidis, G., Bezerianos, A., Dimarogonas, A. and Sutherland, C. (1990) 'Monitoring of fracture healing by lateral and axial vibration analysis', *Journal of Biomechanics*, 23(4), pp. 323–330.

[3] <https://www.bksv.com/media/doc/bv0013.pdf>

[4] <https://dl.acm.org/doi/10.1145/3097983.3097997>

## 5 Legal, Safety and Regulatory Considerations

### 5.1 Intended use and scope of the device

The regulation of medical devices is influenced by intended use, which determines whether a product qualifies as a medical device and the level of regulatory oversight required. In the UK, the Medicines and Healthcare products Regulatory Agency (MHRA) defines a medical device as “any instrument, apparatus, appliance, software, implant, reagent, material or other article intended by the manufacturer to be used alone or in combination, for human beings” [1].

The proposed device is a non-invasive medical device intended for monitoring fracture healing over time. Its primary function is to collect and analyse biomechanical sensor-based data to track changes in bone healing progression. The integrated artificial intelligence (AI) component is used exclusively for trend recognition and identifying patterns or deviations from expected healing trajectories. Importantly, the AI does not make autonomous clinical decisions and does not provide a diagnosis for the patient. All clinical interpretation and decision-making remain under the responsibility of a qualified healthcare professional.

[1] Medicines and Healthcare products Regulatory Agency (2020). *Regulating Medical Devices in the UK*. [online] GOV.UK. Available at: <https://www.gov.uk/guidance/regulating-medical-devices-in-the-uk>.

### 5.2 Medical device regulation and classification

Medical devices are regulated by the MHRA under the UK framework. Devices must be registered with the MHRA and meet conformity requirements before being placed on the market, with compliance demonstrated through the UK Conformity Assessed (UKCA) marking [2][3]. Based on risk-based classification principles and comparison with non-invasive, battery-powered devices such as TENS units [4], the proposed fracture monitoring device would likely be classified as Class IIa. This classification would require assessment by a UK Approved Body and ongoing post-market surveillance, including adverse incident reporting to the MHRA.

[2] Department of Health and Social Care (2023) *Regulating medical devices in the UK*. Available at: <https://www.gov.uk/guidance/regulating-medical-devices-in-the-uk>.

[3] Department of Health and Social Care (2023) *Using the UKCA marking*. Available at: <https://www.gov.uk/guidance/using-the-ukca-marking>.

[4] Cleveland Clinic (2023) *Transcutaneous electrical nerve stimulation (TENS)*. Available at: <https://my.clevelandclinic.org/health/treatments/15840-transcutaneous-electrical-nerve-stimulation-tens>.

### 5.3 Regulatory pathway and market selection

The initial market is the UK National Health service (NHS), with deployment limited to Great Britain (England, Scotland, and Wales), meeting with MHRA and UKCA requirements. At this stage,

regulatory considerations outside the UK (such as the US FDA clearance and EU CE marking under MDR) are not prioritised.

## 5.4 AI and Software as a Medical Device (SaMD)

The AI component of the device falls under the scope of Software as a Medical Device (SaMD), as it performs a medical function by analysing patient data for a defined clinical purpose [5]. MHRA guidance highlights that the intended purpose of software is key in determining its regulatory classification, level of risk and the evidence required to support it. This means that the function of the AI must be clearly defined and not overstated [5].

In this device, AI is used solely for trend recognition, analysing vibration-based measurements over time to identify changes in bone stiffness and healing progression. The model compares each patient's data with expected recovery patterns, considering factors such as age, sex, and bone type. Importantly, the AI does not diagnose fractures, classify healing stages or make independent clinical decisions. All outputs are presented as supportive trend indicators, with clinicians retaining full responsibility for interpretation and decision-making.

Limiting the AI to non-diagnostic trend analysis reduces the regulatory burden, as the system functions as a decision-support tool rather than a diagnostic algorithm. However, SaMD requirements still apply, including the need for transparency, appropriate validation across patient groups, and measures to prevent over-reliance on AI outputs. The intended purpose of the software must clearly reflect these limitations to ensure compliance with MHRA guidance [5].

[5] Medicines and Healthcare products Regulatory Agency (2023) *Crafting an intended purpose in the context of software as a medical device (SaMD)*. Available at: <https://www.gov.uk/government/publications/crafting-an-intended-purpose-in-the-context-of-software-as-a-medical-device-samd>

## 5.5 Data protection and cyber security

The proposed device will be used in clinical environments, with measurement results transmitted through hospital networks to healthcare professionals. Although the device mainly displays results that are reviewed separately by the clinician, appropriate technical and organisational measures are still required to protect patient information. Under UK data protection principles, organisations must ensure that personal data is protected against unauthorised access, loss, or disclosure (ICO, 2023a). When data is transmitted through hospital systems, secure communication methods such as encryption can be used to reduce the risk of interception (ICO, 2023b). In addition, limiting identifiable information and applying anonymisation or pseudonymisation where possible can help reduce privacy risks (ICO, 2025). Healthcare organisations must also have procedures in place to detect and respond to potential personal data breaches, including incident investigation and reporting where necessary (ICO, 2023c).

For this device, the data being processed remotely includes accelerometer-derived vibration signals, frequency response features, cuff pressure readings and AI-generated healing trends. Although these

measurements do not contain direct identifiers, they are still linked to a patient's condition and are therefore treated as personal data. The use of wireless communication (Bluetooth) between the brace and hospital systems also introduces additional cybersecurity considerations, such as secure pairing, encrypted data transfer and limiting access within clinical networks. Local pre-processing of the vibration signals, for example using Fourier analysis, can also help reduce privacy risks by limiting how much raw data needs to be transmitted.

- a) ICO (2023). *A Guide to Data Security*. [online] ico.org.uk. Available at: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/security/a-guide-to-data-security/>.
- b) ICO (2023). *Encryption*. [online] Available at: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/security/encryption/>.
- c) ICO (2023). *Personal Data breaches: a Guide*. [online] ico.org.uk. Available at: <https://ico.org.uk/for-organisations/report-a-breach/personal-data-breach/personal-data-breaches-a-guide/>.
- d) ICO (2025). *Anonymisation*. [online] Available at: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-sharing/anonymisation/>.

## 5.6 Ethical considerations

The development and potential clinical use of any proposed device must consider several ethical principles, particularly patient safety, responsible research practice and the protection of participant rights. Ensuring that medical devices are safe and effective for their intended use is a key ethical responsibility during development, and devices should be carefully evaluated before they are used more widely in clinical settings (Herth, Maldonado and Brock, 2024).

For this device, ethical considerations are linked to its non-invasive vibration-based measurement method, the use of an inflatable cuff, and the incorporation of AI for analysing healing trends. Although the device avoids radiation exposure and reduces the need for repeated X-rays, there are still potential risks that must be considered, such as discomfort from vibration, excessive cuff pressure, or incorrect interpretation of the results. Participants should also be informed that the AI does not provide a diagnosis but instead supports clinicians by identifying trends in fracture healing.

In order to test the device in clinical studies using human participants, ethical approval is required from a Research Ethics Committee (REC) within the UK Health Departments' Research Ethics Service. Applications for medical device research are typically submitted to a flagged REC with relevant expertise (HRA medical devices guidance). The REC would review the study protocols to ensure that participant safety, informed consent and ethical standards are maintained throughout the research process (Standard Operating Procedures for Research Ethics Committees). Ethical assessments must also consider potential risks to participants and whether the expected benefits of the research outweigh those risks (GOV UK, 2023 Approval standards and guidelines). For example, participants would need to be fully informed about how the device monitors fracture healing over time and how their data may be used during the study.

Herth, F.J.F., Maldonado, F. and Brock, J. (2024). Ethics in Medical Device Development. *Respiration*, [online] pp.736–738. doi:<https://doi.org/10.1159/000541963>

Health Research Authority. (n.d.). *Medical devices and software applications*. [online] Available at: <https://www.hra.nhs.uk/planning-and-improving-research/policies-standards-legislation/medical-devices-and-software-applications/>.

Standard Operating Procedures for Research Ethics Committees STANDARD OPERATING PROCEDURES. (n.d.). Available at: [https://s3.eu-west-2.amazonaws.com/www.hra.nhs.uk/media/documents/REC\\_Standard\\_Operating\\_Procedures\\_Version\\_7.7.pdf](https://s3.eu-west-2.amazonaws.com/www.hra.nhs.uk/media/documents/REC_Standard_Operating_Procedures_Version_7.7.pdf).

Gov.UK (2023). *Approval Standards and guidelines: Ethical Assessment*. [online] GOV.UK. Available at: <https://www.gov.uk/government/publications/accessing-ukhsa-protected-data/approval-standards-and-guidelines-ethical-assessment>.

## 5.7 Intellectual Property and Freedom to Operate

Intellectual property must be considered when developing new medical devices. Patents give the owner the legal right to prevent others from making, using or selling the invention for a limited period, usually around twenty years (WIPO, n.d.). Because of this, it is important to check whether similar technologies already exist. A preliminary patent search identified a relevant patent titled *Method and system for assessing the state of healing of a fractured long bone* (Chiu, Russ and Fitzgerald, 2019). This patent describes a system that assesses fracture healing using vibration measurements, similar to the concept introduced in the research that informed the design of this device (Vien et al., 2022). However, the proposed OsetoTrack device differs in its implementation by using a vibration motor to generate a frequency sweep, from which monitoring occurs externally with two accelerometers as compared to invasively implanted sensors. Although we consider this concept to be distinct from existing work, a full Freedom to Operate analysis would be required before commercial development to confirm no patent infringement.

World Intellectual Property Organization (WIPO) (n.d.) *Patents*. Available at: <https://www.wipo.int/en/web/business/patents>

Chiu, W., Russ, M. and Fitzgerald, M. (2019) *Method and system for assessing the state of healing of a fractured long bone*. Australian Patent AU2019900018.

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## 5.8 Electrical safety standards

Electrical safety for medical electrical equipment is governed by BS EN 60601-1:2006, which specifies requirements for basic safety and essential performance (British Standards Institution, 2006). The proposed fracture monitoring device includes several electrical components, including a vibration motor used to generate mechanical excitation, two accelerometers for sensing vibration

transmission, signal-processing electronics (ADC, amplifiers and filters), a Li-ion battery supply with a battery management system, and a Bluetooth communication module. The sensing system and inflatable cuff contact the patient’s skin, meaning the device contains patient-contact components and would likely be classified as a Type BF applied part under this standard.

At the concept stage, the standard highlights key safety considerations including electrical isolation between sensing circuits and power electronics, limitation of leakage currents from patient-contact sensors, prevention of overheating of electronic components and battery systems, and ensuring electromagnetic compatibility of the Bluetooth communication module.

See Appendix 3: Electrical Safety Considerations (BS EN 60601-1:2006) for a full review of the electrical safety considerations.

### 5.9 Risk assessment and mitigation

Risk management is an important aspect of medical device development and follows the international standard BS EN ISO 14971:2019, which outlines how hazards associated with medical devices should be identified, evaluated and controlled (British Standards Institution, 2019).

For the proposed fracture monitoring device, potential risks may arise from the electrical system, vibration motor, inflatable cuff used to stabilise the limb, and AI-based signal analysis. Table 3 summarises key hazards identified at the concept stage and the measures considered to reduce these risks, including electrical insulation, battery protection systems, pressure limits for the cuff and clinician oversight of AI-generated trends.

Table 3. Example Risks and Mitigation Strategies (BS EN ISO 14971:2019)

| Hazard                         | Potential Harm                    | Mitigation                                                                                                      |
|--------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Electrical fault               | Shock or device malfunction       | Electrical insulation and compliance with safety standards (e.g. isolation between sensing and motor circuits)  |
| Battery overheating            | Burn or fire risk                 | Battery management system (BMS) i.e. small electronic circuit that protects and controls a rechargeable battery |
| Excessive vibration            | Patient discomfort                | Controlled vibration frequency and motor limits                                                                 |
| Excessive cuff pressure        | Circulatory restriction           | Pressure sensor and pressure limits                                                                             |
| Misinterpretation of AI output | Incorrect clinical interpretation | AI limited to trend analysis only and not diagnosis (done by clinicians)                                        |

British Standards Institution (2019) BS EN ISO 14971:2019 Medical devices – Application of risk management to medical devices. London: BSI.

## 5.10 Clinical trials

Clinical evidence is often required to demonstrate that a medical device is safe and performs as intended. This evidence may come from published literature, data from equivalent devices, or from a clinical investigation involving human participants. Clinical investigations are typically required when sufficient clinical evidence is not already available (NHS, 2020).

In the UK, clinical research involving medical devices is governed by national medical device regulations and research governance frameworks. Besides the ethical approval already discussed, clinical investigations must receive approval from the Medicines and Healthcare products Regulatory Agency (MHRA). Health Research Authority (HRA) approval is also required for research involving NHS patients or their data (Medicines and Healthcare products Regulatory Agency, 2014).

For this device, a clinical investigation would focus on both safety and performance of the vibration-based monitoring approach. Fracture patients would periodically be measured during their recovery, with vibration measurements collected over time to track changes in bone stiffness. These measurements would then be compared with standard clinical assessments, such as X-rays or clinician-reported healing outcomes, to determine whether the device can reliably monitor healing progression. As part of the study design, X-rays would be limited to standard clinical imaging schedules to avoid unnecessary radiation exposure, with the device ultimately aiming to reduce the need for repeated imaging over time.

Performance would be assessed by checking that the device produces consistent measurements, can detect expected changes in bone stiffness during healing, and shows agreement with clinical outcomes. Safety would focus on ensuring that repeated vibration does not cause discomfort, that cuff pressure remains within safe limits, and that the electrical system (including the battery and sensors) operates safely during use. As the device includes AI, the study would also need to confirm that the trend outputs are reliable across different patient groups and that clinicians do not rely on them as a diagnosis. The intended use must remain clearly limited to decision support.

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## 6 Market Research

### 6.1 Target Customers

OsteoTrack is aimed at a clear customer group – NHS trauma and orthopaedic services managing fracture follow-up, with patients recovering from long-bone fractures as the end users. These patients represent a population with an unmet need for more accessible and continuous monitoring of healing progress that reduces dependence on ionising radiation and specialist hospital services. Positioning NHS clinics as the buyer rather than individual patients is strategically important for two reasons: firstly, NHS procurement is a far more stable and scalable route to market than direct-to-patient sales as a single trust-level adoption decision reaches thousands of patients annually, as compared to the unpredictable purchasing behaviour of individual patients. Secondly, the core value of our device is not only its benefit to the patient, but also its ability to support more cost-effective fracture follow-up care and lowering the downstream expenses of non-union. Hence, our product would be more suitable as a healthcare system investment than a direct patient purchase.

### 6.2 Market Need and Size

In the UK, there are approximately 850,000 new fractures each year [1] and up to 9% progress to non-union [1]. This creates a substantial burden for the NHS since treating a single non-union case is estimated to cost between £7000 and £79,000 in direct healthcare expenditure [1-2]. Earlier detections of non-union could therefore translate into meaningful cost savings by supporting improved recovery outcomes and reduced follow-up.

The market need for OsteoTrack arises from the current inefficiency in monitoring of fracture healing. Following standard fracture clinic follow-up in the NHS fracture healing pathway for long bones, many patients require serial X-rays and repeated clinic reviews over several weeks or months [3], especially if healing is delayed or non-union is suspected. For non-union assessment, NHS guidance commonly involves at least two radiological reviews about four weeks apart [4]. Consequently, the pathway can involve multiple hospital visits and repeated imaging before healing is confirmed. While essential for diagnosis, this traditional protocol is heavily reliant on X-rays which results in patients being exposed to cumulative ionising radiation that may be unnecessary in some cases, while providing limited added value in later stages of recovery and no evidence of improving early detection. Our device directly addresses this gap by replacing limited X-ray based follow-ups with frequent, radiation-free monitoring administered at a local point of care rather than a centralised fracture unit within a hospital.

OsteoTrack sits at the intersection of two growing markets: AI-supported diagnostics and wearable medical devices. The rapid expansion of the AI in healthcare market in the UK, which is forecasted to grow at a compound annual growth rate (CAGR) of 24.4% in the next 8 years<sup>5</sup> [5], and the projected

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<sup>5</sup> The UK AI in Healthcare market is valued at £316.75 million in 2025 and is projected to reach £2.25 billion by 2034.

26.3% CAGR from 2024 to 2030 for wearable medical devices<sup>6</sup> [6] suggest a favourable market environment for our solution. Furthermore, the adoption of VFCs demonstrates that the NHS is shifting towards reducing pressure on specialist fracture units. This direction of service delivery aligns well with the rationale behind our device, designed to bring fracture-healing assessment closer to patients. Together, these trends reinforce that our OsteoTrack is well-positioned to enter a healthcare landscape with an established appetite for alternative fracture care solutions.

## 6.3 Competition

The competitive landscape for the monitoring of bone fracture healing can be divided into three categories: non-invasive vibration-based systems [7-8], invasive instrumented implants [9-11], and emerging optical sensing approaches [12].

Vibration-based methods, such as two-sensor monitoring [7] and impact testing [8], have been demonstrated to quantitatively track femur healing through frequency response changes. While technically similar to our device in sensing principle, both require clinical specialists to operate and take measurements, a dependency eliminated by OsteoTrack. Next, invasive instrumented implants, like smart bone plates [9], bioelectronic osteosynthesis plates [10], and the Fracture Monitor [11] avoid many of the signal-attenuation challenges of non-invasive external measurements by acquiring data directly invasive implantation at the fracture site or fixation environment. Finally, optical sensing represents an emerging alternative for monitoring fracture healing: laser Doppler and white light spectroscopy have demonstrated the ability to distinguish non-union early through oxygen saturation differences [12]. However, taking these measurements would require precise probe placement over the fracture gap by a trained clinician in a specialist setting.

While each existing competitor addresses parts of the clinical gap left by X-ray based monitoring, they do not necessarily reduce the complexity or cost of the current follow-up pathway. OsteoTrack ultimately differentiates itself through simplicity in measurement and AI integration to automate signal processing, reducing the reliance on specialists and enabling fracture follow-up to be decentralised from specialist hospital settings.

## 6.4 Pricing Model

The company will adopt a per-patient pricing model, under which the NHS pays a fixed fee (see Pricing Structure) for each patient monitored across their full fracture-healing episode, while the device is owned and maintained by the company. This model is chosen over a per-device or capital purchase model due to its lower barrier to adoption. NHS Trusts and GPs can evaluate the device without committing to high upfront capital expenditure, making it particularly attractive for pilot studies and early deployment. Moreover, this aligns with established NHS per-use reimbursement precedents, such as remote cardiac monitoring [13-14] and continuous glucose monitoring [15-16], both of which are funded on a per-patient basis. From a commercial perspective, the model offers

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<sup>6</sup> The UK wearable devices market is valued at around £906.8 million in 2023.

stronger revenue scalability, with income growing in proportion to patient usage, compared to a per device model where revenue would be capped after initial device sales.

A key limitation of the per patient model is the risk of underuse, given that revenue will be directly affected if clinical uptake is lower than expected. This may occur during early implementation if the device is not yet consistently incorporated into routine fracture follow-up or if staff training is limited and incomplete. Nevertheless, this risk is considerably less severe than the fundamental commercial risk associated with a per-device model caused by slow or delayed adoption.

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## 7 Finance

### 7.1 Cost of Goods

The components of OsteoTrack have been priced up as shown in Table 4 (the source for each can be found in Appendix 1: Components). Where bulk discounts are available, these are reflected in the per unit pricing, and where the sales price is denominated in a currency other than GBP, the price has been converted to GBP at the appropriate foreign exchange rate. Prices and exchange rate are effective as of February 2026. For each component, the focus has been on sourcing an item which has at least the minimum necessary characteristics for its purpose (e.g. size, sensitivity, robustness etc. as appropriate) at the cheapest price commercially available. The most expensive are the motor and the accelerometers, both of which must be of sufficient quality to be both durable and able to deliver/sense the desired frequency range.

*Table 4. Component List, Purpose, and Cost Breakdown per Unit*

| Item                                                   | Purpose                      | Quantity for 1 unit | Price (£ inc. VAT) per item |
|--------------------------------------------------------|------------------------------|---------------------|-----------------------------|
| 3D printer filament (100g per unit): Motor Shaft       | Frequency production         | 1                   | 5.46                        |
| Stainless steel disc 20mm diameter x 1.5mm thickness   | Frequency production         | 1                   | 0.12                        |
| Cylindrical Brushless Motor                            | Frequency production         | 1                   | 108.32                      |
| Motor Control                                          | Frequency production         | 1                   | 11.58                       |
| Accelerometer                                          | Frequency Sensing            | 2                   | 84.82                       |
| Lithium-Ion Rechargeable Battery Pack                  | Power                        | 1                   | 25.99                       |
| Battery Management System                              | Power                        | 1                   | 3.49                        |
| Cabling (1m per device)                                | Power                        | 1                   | 1.14                        |
| Blood Pressure Cuff with pump                          | Device housing & positioning | 1                   | 8.20                        |
| 3D printer filament (100g per unit): Rectangular strip | Device housing & positioning | 1                   | 1.99                        |
| Wireless Bluetooth Module                              | Interface                    | 1                   | 3.28                        |
| Subtotal per unit:                                     |                              |                     | 339.21                      |
| Margin for delivery costs                              | 5%                           |                     | 16.96                       |
| Materials Cost per unit:                               |                              |                     | 356.17                      |

Manufacturing is assumed to be done on UCL campus by the 6 founding members of Osteotrack. Excluding labour costs of time spent assembling the units, and assuming access to a Bambu 3D

printer (available at UCL), the cost of manufacturing 100 units of the device is **£35,617** (see Fundraising Plan for the options to raise this sum). Including labour costs, this rises to £40,117.

The assumed lifetime of OsteoTrack is estimated at 5 years. This takes into account anticipated advances in technology and improvements prompted by ongoing feedback from patients/users, as well as anticipated durability of the constituent parts.

## 7.2 Pricing Structure

As described in Pricing Model 6.4, the intention is to adopt a fee-per-patient pricing structure, with the device to be provided by the manufacturing company (and remain the property of the manufacturing company); under this arrangement, the device may be used as many times as necessary on a single patient. To price appropriately, it is assumed that the technology will initially focus on upper limb injuries (humerus, ulna, radius), which occur at a rate of approximately 303 per 100,000 people per annum<sup>7</sup>, which equates to just over 200,000 per annum in the UK. Aiming for a market penetration of 50% (i.e. stability level of 100,000 patients treated per annum) with a circulation of 4,000 units once ramp-up has been achieved, we anticipate a fee-per-patient of £35. This is significantly cheaper than a single upper limb X-ray cost (~£56, as noted in

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<sup>7</sup> <https://pubmed.ncbi.nlm.nih.gov/38544581/>

Clinical Background) and, of course, has the advantage of multiple uses per patient, with no associated harmful ionising radiation. It is anticipated that for most fractions, additional X-rays after the initial diagnostic X-ray would be unnecessary, and that early warning of potential non-union would allow the more complex cases to be identified and treated earlier, which is a further cost saving (see

Clinical Background for the costs associated with complex non-union cases).

Once successful implementation of OsteoTrack has been established and recognised within the medical community, it is anticipated that use would be extended to lower limb injuries also, thus expanding the patient pool significantly.

## 7.3 Profit & Loss and Cashflow Projections

Appendices 1 and 2 contain detailed ten-year annual Profit & Loss and Cashflow projections for a small company created for the purpose of manufacturing and maintaining the devices and associated back-end technology, advertising and distribution, adhering to all necessary regulations (see Legal, Safety and Regulatory Considerations), and providing equipment, office/lab space and salaries to required employees.

In these projections, it is anticipated that it will take two years to conduct a successful clinical trial and gain all the necessary approvals. Following that initial period, ramp-up to stabilised use levels is assumed to take a further 5 years.

The most significant Profit & Loss line items are the following:

- **R&D:** Clinical trials to prove the efficacy of the device and its utility in replacing additional (post-diagnosis) X-rays for common fractures are anticipated, at a total cost of £315,000 (spread over two years); this is based on a clinical trial of 30 patients at £10,000 per patient<sup>3</sup>, with £15,000 being the cost of attaining regulatory approval. This is by far the most significant line item but will be vital for the success of the device.
- **Marketing:** This device will be marketed direct to the NHS, and therefore distribution costs will not be incurred, but significant marketing expenses (at 15% of gross revenue) have been included. These expenses will cover the costs associated with conferences, key decision-maker engagement, marketing material and a specialist sales team (see Marketing Strategy and Fundraising Plan for more detail).
- **Salaries:** The six members of the founding OsteoTrack team are assumed to work part-time (0.5 of a week) on the project for the first 4 years (through the initial two year trial period and the first two years of ramp-up) alongside a full-time software engineer. After this time, the founding team will reduce to 3 people working part-time alongside a part-time software engineer.
- **Financing Costs:** It is expected that funding will come from a combination of equity contributions from the founding OsteoTrack team and debt/equity from commercially interested parties. The projections in Appendix 2 assume debt financing.

## 8 Marketing Strategy and Fundraising Plan

### 8.1 Marketing strategy

To reinforce the presence of the device as an orthopaedic monitoring platform and not as a consumer wearable, professional branding is necessary. The key selling point of this device is that it monitors fracture healing over time without exposure to harmful ionising radiation; to reflect this, and to reinforce orthopaedic use, the name chosen is OsteoTrack. AI is used by OsteoTrack for interpretation, and the technology should be used to enhance, rather than replace, clinical judgement; it will be critical that the marketing campaign highlights this point for high clinical adoption.

The communications strategy should identify and target influential stakeholders. Although the general public is the end user of NHS technologies, the decision-makers on whether to adopt the device will be clinical and institutional professionals who recognise that OsteoTrack will improve workflow, reduce unnecessary follow-up, enable safer discharge decisions, and provide meaningful clinical value. As such, the main audiences for OsteoTrack are orthopaedic consultants and VFC leads as well as NHS procurement bodies at point of purchase. Thus, the marketing strategy should be conceived primarily as a business-to-clinician and business-to-institution communications strategy, not a consumer advertising campaign.

The core marketing message emphasizes three pillars: cost-reduction, workflow efficiencies and alignment with remote care pathways (as detailed elsewhere in this report). This message is consistent with the larger NHS effort directing focus to digital and remote monitoring pathways in fracture care.

An effective way to enter the market is through private orthopaedic clinics, while targeting the NHS as the long-term strategic market. The private clinics have a shorter procurement cycle and more flexibility in piloting innovative technology and gathering of early case studies and implementation feedback. These initial deployments can serve as a stepping stone to NHS adoption by providing real-world examples of workflow enhancement and safety. Simultaneously, the device is particularly well adapted to VFCs which already rely on remote review models and are pressurised to eliminate unnecessary facetime appointments. OsteoTrack fits this model as a tool to allow clinicians to assess the healing progression at local clinics and decide if further investigation is required.

It should be stressed to clinicians that all communications to the patient must remain clinically responsible. For example, clinicians should not display device data to patients in a way that is likely to cause anxiety or promote the product as an autonomous diagnostic system. The device is intended to give clinicians better longitudinal insight which will in turn provide patients with more reassurance.

The best channels for outreach to decision-makers are conference presentations, clinical white papers, professional society engagement and innovation programme participation. This should be done through the British Orthopaedic Association Congress, the British Orthopaedic Research

Society, white paper publication, and applications to the NHS Innovation Accelerator. The strength of this approach is that it puts OsteoTrack directly in the hands of the clinicians, researchers and other decision-makers most likely to support adoption. Early promotion should facilitate a private pilot in 2-3 clinics with 30-50 patients. Following this, a contained NHS pilot in one Trust using 100 devices should be activated. The staged marketing activity demonstrates realism, reduces risk of failure in the market, and produces the evidence base required for later procurement discussions.

See <https://osteo-track-flow.base44.app> for an early demonstration of the digital presence envisaged for OsteoTrack.

## 8.2 Fundraising Plan

As a regulated medical device venture rather than a low-cost software startup, OsteoTrack will require external capital. The funding requirement extends beyond simple production costs and must also support intellectual property protection, prototype refinement, quality management systems and early clinical validation. Current project estimates place the per-device production cost at £401.17 including labour, meaning that manufacturing 100 units would require approximately £40,170 (see Cost of Goods). When patent costs, staffing, equipment, quality management systems, and clinical trials are included, the estimated startup capital rises to approximately £1,135,000 for a long-term viable business plan (see Appendix 12.1 and 12.2). This makes it clear that the venture cannot rely on casual self-funding, and that a structured fundraising strategy is necessary from the outset.

The most appropriate fundraising strategy is to follow a grants-first, equity/debt-later model. Currently, OsteoTrack is still progressing from academic feasibility toward proof of concept, so the strongest initial pathways are university translational grants, early-stage health innovation grants, and government-backed MedTech funding schemes. For example, the UCL Inspire Innovation Fund (Health) typically funds projects in the range of £20,000 to £80,000, which makes it suitable for prototype refinement and short translational studies [1]. Similarly, the NIHR i4i FAST scheme provides £50,000 to £100,000 for 6–12 month projects, making it well matched to feasibility-stage technical and clinical development [2]. These non-dilutive routes are especially attractive because they allow the company to strengthen evidence and refine the prototype without immediate loss of equity.

Once OsteoTrack has progressed beyond initial feasibility, larger translational funding routes become more appropriate. The NIHR i4i Product Development Awards are particularly relevant for medical technologies, with funding uncapped in principle, although awards typically fall in the range of £500,000 to £1.5 million [3]. Likewise, the Biomedical Catalyst offers a strong UK health innovation route: current published guidance indicates that small projects can receive up to £500,000, while large projects can receive up to £2 million [4]. These schemes are attractive because they are specifically designed for healthcare innovation and therefore align well with a fracture-monitoring device that must demonstrate both technical function and clinical relevance.

A broader alternative is Innovate UK Smart Grants, which remain relevant where OsteoTrack is presented as a commercially viable innovation project with wider economic impact. The latest published Smart Grant guidance indicates that 6–18 month projects must have total costs of £100,000–£500,000, while 19–24 month projects must have total costs of £100,000–£1 million, with up to 70% of eligible costs covered depending on business size and project type [5]. For OsteoTrack, this makes Smart Grants a useful option if the company can show not only technical novelty but also a clear commercial route to market.

In addition to grants, OsteoTrack could benefit from accelerator-based support. The MedTech SuperConnector / MSK Innovation Accelerator offers more than £80,000 worth of academic and commercialisation support, together with the opportunity to pitch for up to £100,000 investment [6]. This type of support is particularly useful for investor readiness, strategic mentorship, and early commercial development. Another pathway is Entrepreneurs First, which offers an equity-free founder grant and states that qualifying teams may receive up to \$250,000 investment [7]. However, these accelerator pathways should be seen as complements to translational grants rather than replacements for them, since OsteoTrack's early needs are still dominated by technical validation and clinical evidence generation rather than rapid venture scaling.

The fundraising plan should therefore be tied to a sequence of milestones:

- (i) OsteoTrack team should secure provisional IP protection and university technology transfer support
- (ii) The team should pursue early non-dilutive funding through UCL translational support or NIHR i4i FAST to strengthen the prototype and collect initial evidence [1], [2]
- (iii) Once proof of concept is largely established, the company should apply for larger-scale translational funding such as the Biomedical Catalyst or i4i PDA to support pilot readiness and more formal clinical validation [3], [4]
- (iv) Only after those steps should OsteoTrack approach seed or angel investment or lenders, since equity/debt finance is likely to become more achievable once technical risk and translational uncertainty have already been reduced

This plan is attractive because it demonstrates disciplined capital use and shows that each funding stage is linked to a specific reduction in technical and commercial risk.

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## 9 Project Risks and Mitigants

The main risks and any relevant mitigants associated with this project are detailed in Table 5.

Table 5. Risks and Mitigations of the Project

| Risk                                                                                                                                  | Mitigant                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Targeted market is one single healthcare provider (NHS) and thus the business plan relies heavily on adoption throughout the NHS      | The NHS is one of the largest healthcare providers globally. Whilst adoption throughout the NHS is the goal, initial use by a small number of clinics will provide sufficient data and evidence to entice other centres to participate.                                                                                                                    |
| Clinic trials may be more expensive and take longer than initially expected                                                           | The founding team would work with the financing partnership to address this possibility. Worst case scenarios can be produced and provided to prospective finance partners. If higher costs are incurred, the anticipated price per patient (£35) is extremely reasonable and can be increased whilst still providing significant cost savings to the NHS. |
| Unauthorised access to patient data, leading to potential breach of patient confidentiality                                           | Access restricted to authorised healthcare professionals through clinic and hospital network controls                                                                                                                                                                                                                                                      |
| Interception of data during transmission, leading to exposure of sensitive medical information                                        | Use of secure network protocols and encryption during data transfer                                                                                                                                                                                                                                                                                        |
| Patient identification from recorded measurements, leading to regulatory non-compliance with privacy and data protection requirements | Use of anonymisation or pseudonymisation where possible and appropriate                                                                                                                                                                                                                                                                                    |
| Personal data breach, resulting in potential harm to individuals and regulatory non-compliance                                        | Incident detection procedures and breach reporting protocols                                                                                                                                                                                                                                                                                               |
| Repositioning error between appointments                                                                                              | The app could perform a pre-sweep check using the live signals from both accelerometers. If placement is incorrect, the relative signal between the two sensors will fall outside an expected range, prompting the GP to reposition before measurement begins.                                                                                             |

|                                                                |                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cuff inflation may cause pain or displace an unstable fracture | If trials show consistent discomfort, the target pressure range can be reduced provided signal quality remains sufficient.                                                                                                                              |
| Motor vibration captured in accelerometer recording            | Vibrational dampening material in the device; Disabling the motor when recording data.                                                                                                                                                                  |
| AI model has different accuracy for different demographics     | Private clinic deployment for diverse data collection and clinical trials for validation                                                                                                                                                                |
| Insufficient training data                                     | Parallel deployment in private orthopaedic clinics, data augmentation techniques, increasing number of frequencies included in the T-LSTM input                                                                                                         |
| Poor model prediction ability                                  | Inclusion of additional variables, including patient demographics and initial X-ray image. Simplification of frequency response input data through use of CNN or the DBSCAN algorithm. Data augmentation, regularisation, and normalisation techniques. |

## 10 Conclusion

TBD

## 11 References

TBD

## 12 Appendixes

### 12.1 Appendix 1: Components

| Subsystem  | Component                                 | Price Source                    | Material                  | Specifications                                        | Design justification                                                                                                           |
|------------|-------------------------------------------|---------------------------------|---------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Excitation | Cylindrical brushless DC motor            | <a href="#">RS</a>              | —                         | 12V; operable up to 500Hz                             | 12V matches battery output; 500Hz ceiling far exceed 60-110 Hz target range [1]                                                |
|            | Motor control                             | <a href="#">Amazon</a>          | —                         | PWM control; 12-36 V input                            | Steps the motor through the frequency sweep in precise controlled increments                                                   |
|            | Motor shaft (3D printed)                  | <a href="#">Bambu Store</a>     | PA6-GF nylon, 100% infill | Shaft geometry                                        | Provides the rigidity needed to transmit oscillatory forces; Shaft converts rotation to linear oscillation at the contact disk |
|            | Contact disc                              | <a href="#">Amazon</a>          | 316 stainless steel       | 20 mm diameter; 1.5 mm thick                          | Biocompatible and sterilisable for repeated skin contact                                                                       |
| Sensing    | Accelerometers (x2)                       | <a href="#">Farnell</a>         | —                         | 1-11,000 Hz range; 5 mV/g sensitivity; 3.3-5 V supply | Frequency range covers 60-400 Hz measurement band with margin; low voltage supply good for our battery                         |
|            | Accelerometer mounting strip (3D printed) | <a href="#">Bambu Store</a>     | ABS plastic               | Flexible; adjustable slot spacing                     | Conforms to different limb curvatures; adjustable slots allow repositioning for patient size                                   |
| Housing    | Blood pressure cuff + automated pump      | <a href="#">Brother Medical</a> | —                         | Target inflation: 120-180 mmHg                        | Inflating the cuff simultaneously holds the motor disc and accelerometers firmly in place against the limb, and                |

|           |                             |                                   |   |                                                           |                                                                                                                                                                            |
|-----------|-----------------------------|-----------------------------------|---|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                             |                                   |   |                                                           | compresses overlying soft tissue to reduce signal damping. The validated pressure range of 120–180 mmHg [1] ensures repeatable mechanical coupling regardless of operator. |
| Power     | Rechargeable Li-ion battery | <a href="#">Amazon</a>            | — | 12V                                                       | 12V directly powers the motor; rechargeable for repeated clinical use                                                                                                      |
|           | Battery management system   | <a href="#">Amazon</a>            | — | Over-voltage, under-voltage, and short-circuit protection | Required for electrical safety compliance                                                                                                                                  |
| Interface | Bluetooth low energy module | <a href="#">RS</a>                | — | BLE 5.0; range 10 m                                       | Sufficient range for bedside-to-tablet wireless transmission; low power consumption preserves battery life between charges                                                 |
|           | Cabling                     | <a href="#">Click4Electronics</a> | — | 1.5mm 2 Core 3182Y Black Flexible Cable                   | —                                                                                                                                                                          |

## 12.2 Appendix 2: Financial Assumptions, P&L Forecast and Cashflow Forecast

### Revenue Assumptions (Leasing Model):

|                                     |         |
|-------------------------------------|---------|
| Price per patient (GBP)             | 35      |
| Assumed Lifetime per device (years) | 5       |
| Stabilised Patients per annum       | 100,000 |
| Stabilised Devices in Circulation   | 4,000   |

### Cost of Goods Produced Assumptions:

#### Manufacturing:

|                                     |          |
|-------------------------------------|----------|
| Manufacturing Tools (miscellaneous) | 1,000.00 |
| Bambu 3D printer                    | 2,441.13 |

#### QMS System:

|                         |          |
|-------------------------|----------|
| Number of Users         | 3        |
| Cost per User per month | 1,000.00 |

#### Warranty Reserve:

|                             |    |
|-----------------------------|----|
| Percentage of Gross Revenue | 3% |
|-----------------------------|----|

### Operating Expense Assumptions:

#### Salaries:

| Position          | Full/Part-time (1.0 = FT) | People | Salary pa (FT equivalent) | NICS pa |
|-------------------|---------------------------|--------|---------------------------|---------|
| Software Engineer | 1                         | 1      | 60,000                    | 8,249   |

|                                              |     |   |               |         |
|----------------------------------------------|-----|---|---------------|---------|
| Founding Team (inc. Legal, Reg & Compliance) | 0.5 | 6 | 40,000        | 2,249   |
| Per Annum Total                              |     |   | 180,000       | 21,746  |
|                                              |     |   | Salary + NICS | 201,746 |
| Reduced Team post Year 4:                    |     |   |               |         |
| Software Engineer                            | 0.5 | 1 | 60,000        | 3,749   |
| Founding Team (inc. Legal, Reg & Compliance) | 0.5 | 3 | 40,000        | 2,249   |
| Per Annum Total                              |     |   | 90,000        | 10,498  |
|                                              |     |   | Salary + NICS | 100,498 |

Other Expense Items:

|                                               |         |
|-----------------------------------------------|---------|
| Marketing                                     | 525,000 |
| Rent per annum                                | 14,400  |
| IT (including software licenses) & Accounting | 13,000  |
| Clinical Trials (per annum, Years 1 & 2)      | 157,500 |

Financing Assumptions:

|                          |           |
|--------------------------|-----------|
| Equity Funding from Team | 120,000   |
| Loan Amount              | 1,135,000 |
| Interest Rate            | 10%       |

| Proforma P&L                                 | Year 1           | Year 2           | Year 3           | Year 4           | Year 5           | Year 6           | Year 7           | Year 8           | Year 9           | Year 10          |
|----------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Ramp-up assumed                              | 0.0%             | 0.0%             | 20.0%            | 35%              | 50%              | 70%              | 85%              | 100%             | 100%             | 100%             |
| Patients per annum                           | -                | -                | 20,000           | 35,000           | 50,000           | 70,000           | 85,000           | 100,000          | 100,000          | 100,000          |
| Devices in use                               |                  |                  | 800              | 1,400            | 2,000            | 2,800            | 3,400            | 4,000            | 4,000            | 4,000            |
| Devices manufactured in year                 |                  |                  | 800              | 600              | 600              | 800              | 600              | 1,400            | 1,400            | 2,000            |
| Use payments                                 | -                | -                | 700,000          | 1,225,000        | 1,750,000        | 2,450,000        | 2,975,000        | 3,500,000        | 3,500,000        | 3,500,000        |
| <b>Revenues</b>                              | -                | -                | 700,000          | 1,225,000        | 1,750,000        | 2,450,000        | 2,975,000        | 3,500,000        | 3,500,000        | 3,500,000        |
| <b>Cost of goods provided</b>                |                  |                  |                  |                  |                  |                  |                  |                  |                  |                  |
| Direct Material Costs                        | -                | -                | (284,940)        | (213,705)        | (213,705)        | (284,940)        | (213,705)        | (498,644)        | (498,644)        | (712,349)        |
| Direct Labour Costs                          | -                | -                | (36,000)         | (27,000)         | (27,000)         | (36,000)         | (27,000)         | (63,000)         | (63,000)         | (90,000)         |
| Manufacturing Overhead                       | (344)            | (344)            | (344)            | (344)            | (344)            | (344)            | (344)            | (344)            | (344)            | (344)            |
| 3D Printer Depreciation                      | (244)            | (244)            | (244)            | (244)            | (244)            | (244)            | (244)            | (244)            | (244)            | (244)            |
| Miscellaneous Supplies Depreciation          | (100)            | (100)            | (100)            | (100)            | (100)            | (100)            | (100)            | (100)            | (100)            | (100)            |
| Quality Management System (QMS) Overhead     | (36,000)         | (36,000)         | (36,000)         | (36,000)         | (36,000)         | (36,000)         | (36,000)         | (36,000)         | (36,000)         | (36,000)         |
| Warranty Reserves                            | -                | -                | (21,000)         | (36,750)         | (52,500)         | (73,500)         | (89,250)         | (105,000)        | (105,000)        | (105,000)        |
| <b>Total COGS</b>                            | <b>(36,344)</b>  | <b>(36,344)</b>  | <b>(378,284)</b> | <b>(313,799)</b> | <b>(329,549)</b> | <b>(430,784)</b> | <b>(366,299)</b> | <b>(702,989)</b> | <b>(702,989)</b> | <b>(943,693)</b> |
| <b>Gross Profit</b>                          | <b>(36,344)</b>  | <b>(36,344)</b>  | <b>321,716</b>   | <b>911,201</b>   | <b>1,420,451</b> | <b>2,019,216</b> | <b>2,608,701</b> | <b>2,797,011</b> | <b>2,797,011</b> | <b>2,556,307</b> |
| % of sales                                   | 0%               | 0%               | 46%              | 74%              | 81%              | 82%              | 88%              | 80%              | 80%              | 73%              |
| <b>Selling, General &amp; Administration</b> | <b>(229,146)</b> | <b>(229,146)</b> | <b>(491,646)</b> | <b>(622,896)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> |
| Salaries                                     | (201,746)        | (201,746)        | (201,746)        | (201,746)        | (100,498)        | (100,498)        | (100,498)        | (100,498)        | (100,498)        | (100,498)        |
| Marketing                                    |                  |                  | (262,500)        | (393,750)        | (525,000)        | (525,000)        | (525,000)        | (525,000)        | (525,000)        | (525,000)        |
| Distribution                                 |                  |                  | -                | -                | -                | -                | -                | -                | -                | -                |
| Rent                                         | (14,400)         | (14,400)         | (14,400)         | (14,400)         | (14,400)         | (14,400)         | (14,400)         | (14,400)         | (14,400)         | (14,400)         |
| IT (including software licenses)             | (13,000)         | (13,000)         | (13,000)         | (13,000)         | (13,000)         | (13,000)         | (13,000)         | (13,000)         | (13,000)         | (13,000)         |
| <b>R&amp;D</b>                               | <b>(157,500)</b> | <b>(157,500)</b> | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>-</b>         |
| Clinical Trials                              | (157,500)        | (157,500)        |                  |                  |                  |                  |                  |                  |                  |                  |
| <b>Total Operating Expenses</b>              | <b>(386,646)</b> | <b>(386,646)</b> | <b>(491,646)</b> | <b>(622,896)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> | <b>(652,898)</b> |
| % of revenues                                | 0%               | 0%               | -70%             | -51%             | -37%             | -27%             | -22%             | -19%             | -19%             | -19%             |
| <b>EBITDA</b>                                | <b>(422,746)</b> | <b>(422,746)</b> | <b>(169,685)</b> | <b>288,549</b>   | <b>767,798</b>   | <b>1,366,563</b> | <b>1,956,048</b> | <b>2,144,358</b> | <b>2,144,358</b> | <b>1,903,653</b> |
| % of revenues                                | 0%               | 0%               | -24%             | 24%              | 44%              | 56%              | 66%              | 61%              | 61%              | 54%              |
| <b>Depreciation &amp; Amortisation</b>       | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     | <b>(244)</b>     |
| <b>Operating income</b>                      | <b>(422,990)</b> | <b>(422,990)</b> | <b>(169,930)</b> | <b>288,305</b>   | <b>767,554</b>   | <b>1,366,319</b> | <b>1,955,804</b> | <b>2,144,114</b> | <b>2,144,114</b> | <b>1,903,409</b> |
| % of revenues                                | 0%               | 0%               | -24%             | 24%              | 44%              | 56%              | 66%              | 61%              | 61%              | 54%              |
| <b>Financing expenses</b>                    | <b>(35,000)</b>  | <b>(85,000)</b>  | <b>(113,500)</b> | <b>(113,500)</b> | <b>(100,055)</b> | <b>(49,658)</b>  | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>-</b>         |
| <b>Profit before tax</b>                     | <b>(457,990)</b> | <b>(507,990)</b> | <b>(283,430)</b> | <b>174,805</b>   | <b>667,498</b>   | <b>1,316,660</b> | <b>1,955,804</b> | <b>2,144,114</b> | <b>2,144,114</b> | <b>1,903,409</b> |
| % of revenues                                | 0%               | 0%               | -40%             | 14%              | 38%              | 54%              | 66%              | 61%              | 61%              | 54%              |
| <b>Corporation tax (1)</b>                   | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>(40,701)</b>  | <b>(163,875)</b> | <b>(326,165)</b> | <b>(485,951)</b> | <b>(533,028)</b> | <b>(533,028)</b> | <b>(472,852)</b> |
| <b>Net income</b>                            | <b>(457,990)</b> | <b>(507,990)</b> | <b>(283,430)</b> | <b>134,104</b>   | <b>503,624</b>   | <b>990,495</b>   | <b>1,469,853</b> | <b>1,611,085</b> | <b>1,611,085</b> | <b>1,430,557</b> |
| % of revenues                                | 0%               | 0%               | -40%             | 11%              | 29%              | 40%              | 49%              | 46%              | 46%              | 41%              |

1. UK limited companies pay Corporation Tax (CT) on their profits. The rate depends on the level of profit, with a small profits rate of 19% for profits up to £50,000 and a main rate of 25% for profits over £250,000

| Projected Cashflows                             | Year 1           | Year 2           | Year 3           | Year 4             | Year 5             | Year 6             | Year 7             | Year 8             | Year 9             | Year 10            |
|-------------------------------------------------|------------------|------------------|------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Units Produced                                  | -                | -                | 800              | 600                | 600                | 800                | 600                | 1,400              | 1,400              | 2,000              |
| <b>Income</b>                                   |                  |                  |                  |                    |                    |                    |                    |                    |                    |                    |
| Use Revenue Stream                              | -                | -                | 700,000          | 1,225,000          | 1,750,000          | 2,450,000          | 2,975,000          | 3,500,000          | 3,500,000          | 3,500,000          |
| Financing Income                                | 470,000          | 500,000          | 285,000          | -                  | -                  | -                  | -                  | -                  | -                  | -                  |
| <b>Total Income</b>                             | <b>470,000</b>   | <b>500,000</b>   | <b>985,000</b>   | <b>1,225,000</b>   | <b>1,750,000</b>   | <b>2,450,000</b>   | <b>2,975,000</b>   | <b>3,500,000</b>   | <b>3,500,000</b>   | <b>3,500,000</b>   |
| <b>Expenses</b>                                 |                  |                  |                  |                    |                    |                    |                    |                    |                    |                    |
| COGS                                            | (39,441)         | (36,000)         | (356,940)        | (276,705)          | (276,705)          | (356,940)          | (276,705)          | (597,644)          | (597,644)          | (838,349)          |
| SG&A                                            | (229,146)        | (229,146)        | (491,646)        | (622,896)          | (652,898)          | (652,898)          | (652,898)          | (652,898)          | (652,898)          | (652,898)          |
| R&D                                             | (157,500)        | (157,500)        | -                | -                  | -                  | -                  | -                  | -                  | -                  | -                  |
| Financing Interest                              | (35,000)         | (85,000)         | (113,500)        | (113,500)          | (100,055)          | (49,658)           | -                  | -                  | -                  | -                  |
| Warranty Expenses                               | -                | -                | (21,000)         | (36,750)           | (52,500)           | (73,500)           | (89,250)           | (105,000)          | (105,000)          | (105,000)          |
| <b>Total Expenses</b>                           | <b>(461,087)</b> | <b>(507,646)</b> | <b>(983,085)</b> | <b>(1,049,851)</b> | <b>(1,082,158)</b> | <b>(1,132,996)</b> | <b>(1,018,852)</b> | <b>(1,355,542)</b> | <b>(1,355,542)</b> | <b>(1,596,247)</b> |
| <b>Working Capital Release / (Top-up)</b>       | <b>(8,913)</b>   | <b>7,646</b>     | <b>(1,915)</b>   | <b>-</b>           | <b>-</b>           | <b>-</b>           | <b>-</b>           | <b>-</b>           | <b>-</b>           | <b>-</b>           |
| <b>Net Operating Income before Amortisation</b> | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>175,149</b>     | <b>667,842</b>     | <b>1,317,004</b>   | <b>1,956,148</b>   | <b>2,144,458</b>   | <b>2,144,458</b>   | <b>1,903,753</b>   |
| Amortisation                                    | -                | -                | -                | (134,448)          | (503,968)          | (496,584)          | -                  | -                  | -                  | -                  |
| Tax                                             | -                | -                | -                | (40,701)           | (163,875)          | (326,165)          | (485,951)          | (533,028)          | (533,028)          | (472,852)          |
| <b>Net Operating Income post Amortisation</b>   | <b>-</b>         | <b>-</b>         | <b>-</b>         | <b>-</b>           | <b>-</b>           | <b>494,255</b>     | <b>1,470,197</b>   | <b>1,611,430</b>   | <b>1,611,430</b>   | <b>1,430,901</b>   |



## 12.3 Appendix 3: Electrical Safety Considerations (BS EN 60601-1:2006)

| <b>BS EN 60601-1<br/>Clause</b>                               | <b>Standard<br/>Requirement</b>                                                               | <b>Relevance to<br/>Device</b>                                                    | <b>Design Implications</b>                                                                                                                |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Clause 4.2 – Risk management process                          | Manufacturers must implement a formal risk management process throughout the device lifecycle | Device includes battery supply, vibration motor, sensors and electronic circuitry | Hazards such as electrical faults, overheating and mechanical vibration must be identified and mitigated                                  |
| Clause 6 – Classification of medical electrical equipment     | Equipment must be classified according to protection against electric shock and applied parts | Accelerometers and cuff contact the patient                                       | Device likely classified as Type BF applied part, requiring electrical isolation between patient-connected circuits and power electronics |
| Clause 8 – Protection against electrical hazards              | Limits voltage, current, insulation and circuit separation                                    | Battery powered electronics and motor located near sensing system                 | Isolation required between sensing electronics and motor drive circuitry                                                                  |
| Clause 11 – Protection against excessive temperature and fire | Equipment must prevent overheating and fire hazards                                           | Li-ion battery and vibration motor may generate heat during operation             | Battery management system and thermal protection required                                                                                 |
| Clause 17 – Electromagnetic compatibility                     | Equipment must not interfere with other electrical equipment                                  | Bluetooth module and switching electronics present                                | EMC testing, filtering and shielding required                                                                                             |

British Standards Institution (2006) BS EN 60601-1:2006 Medical electrical equipment – Part 1: General requirements for basic safety and essential performance. London: BSI.

